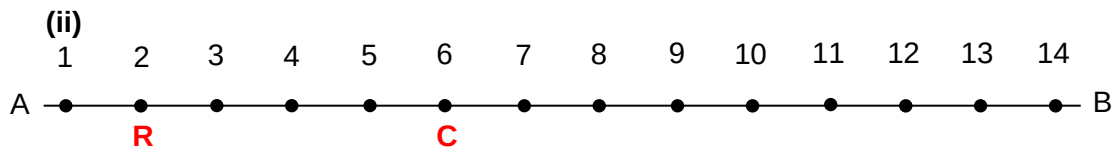


When the rule is balanced, 39% of block M is submerged in liquid.

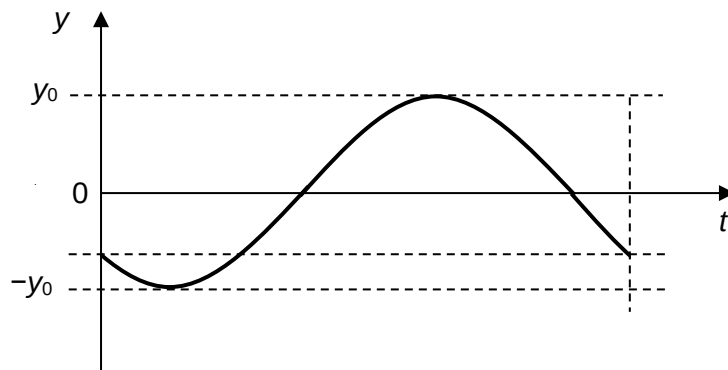
- 4 (a) A longitudinal wave is one in which its particles oscillate in a direction parallel to the direction of energy transfer.

(b) (i) 1. $\frac{1}{2} \times \text{wavelength} = 0.080 \text{ m}$
 $\therefore \text{wavelength} = 0.160 \text{ m}$

2. $\text{frequency} = \frac{\text{speed}}{\text{wavelength}} = \frac{343}{0.160} = 2144 \approx 2140 \text{ Hz}$



(iii)



(iv)

$$I = \frac{P}{4\pi r^2}$$

$$\therefore \frac{I_2}{I_1} = \frac{\cancel{P}/4\pi r_2^2}{\cancel{P}/4\pi r_1^2} = \frac{r_1^2}{r_2^2} = \frac{\left(\cancel{1}/2\right)I}{I}$$

$$r_2^2 = 2r_1^2 \Rightarrow r_2 = \sqrt{2}r_1 = \sqrt{2} \times 0.25 = 0.35355 \text{ m}$$

$$\therefore \text{distance moved by detector} = 0.35355 - 0.25 = 0.10355 \approx 0.10 \text{ m}$$